

Results for Regressions with RHS Proxy

Sep 24, 2024

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Saved ~/Documents/LanguageModel_Labels/congressional_bills/Figures/Estimation/RHS/fig02 t-score vs. l

Saved ~/Documents/LanguageModel_Labels/congressional_bills/Figures/Estimation/RHS/fig03 Bias Density

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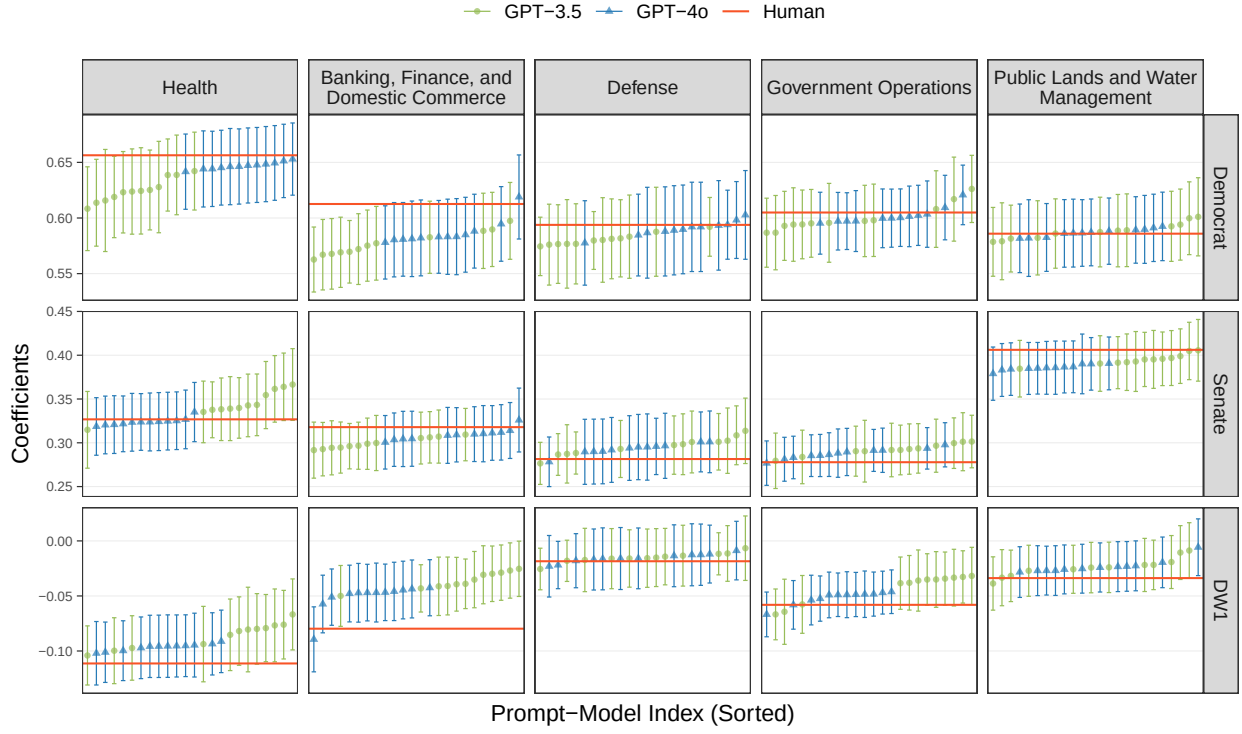
Saved ~/Documents/LanguageModel_Labels/congressional_bills/Figures/Estimation/RHS/fig05 Bias Density

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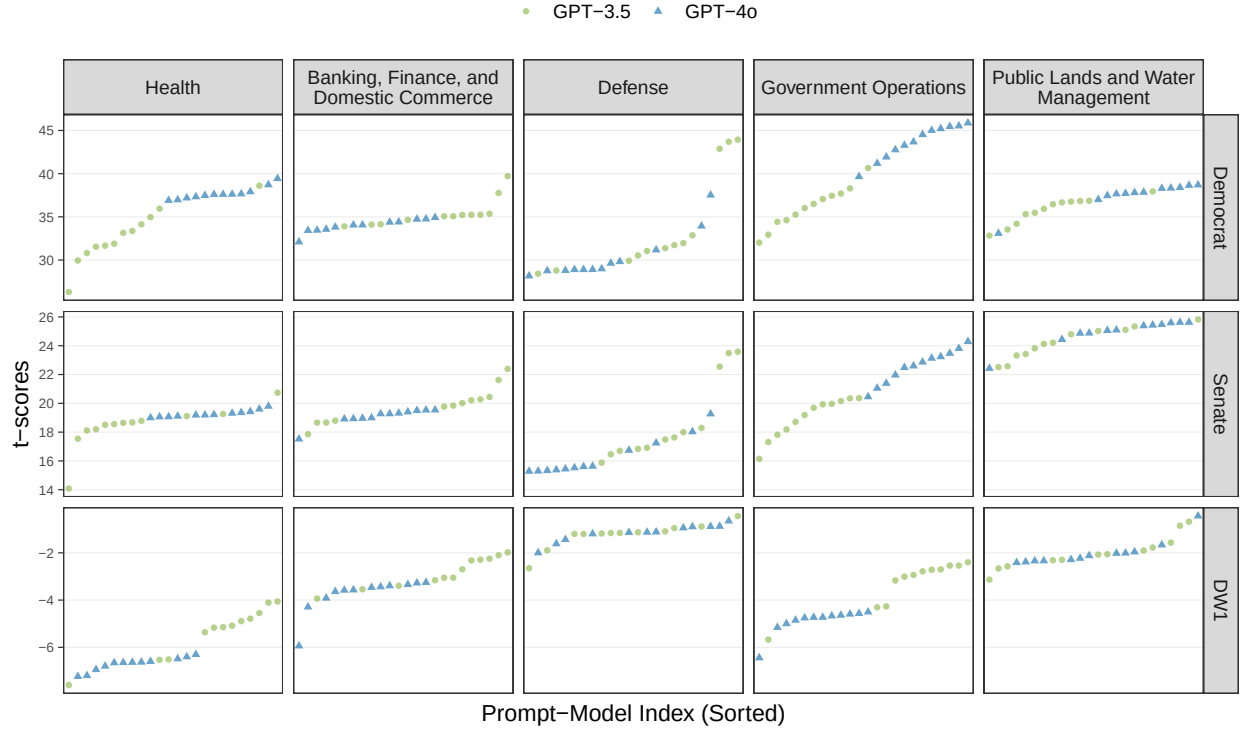
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Figure 1: β_t vs. Prompt for all t . Proxy on the RHS.



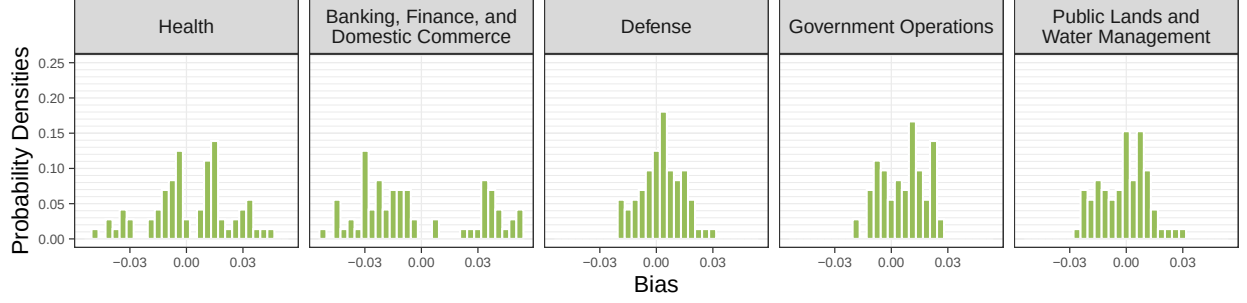
Note: The coefficients were obtained by fitting the model $V = Y^\top \beta = \sum_t Y_t \beta_t$ using 10,000 bills, with 95% confidence intervals computed via robust standard errors. Here, $V \in \{\text{Democrat}; \text{Senate}; \text{DW1}\}$ and $t \in \{\text{Health}; \text{Banking, Finance, and Domestic Commerce}; \text{Defense}; \text{Government Operations}; \text{Public Lands and Water Management}; \text{Other}\}$. The human-based coefficients β^* (red) were estimated using $Y_t^{\text{Human}} := 1\{Y^{\text{Human}} = t\}$, while LLM-based coefficients β^{LLM} (GPT-3.5 in green circles, GPT-4o in blue triangles) used $Y_{t,m,p}^{\text{LLM}} := 1\{Y_{m,p}^{\text{LLM}} = t\}$, with $m \in \{\text{GPT-3.5}, \text{GPT-4o}\}$ denoting the LLM model and $p \in \{1, \dots, 12\}$ representing the prompt. In each subplot, the bias estimates were sorted in ascending order for clarity.

Figure 2: t-scores of each β_t vs. Prompt for all t . Proxy on the RHS.



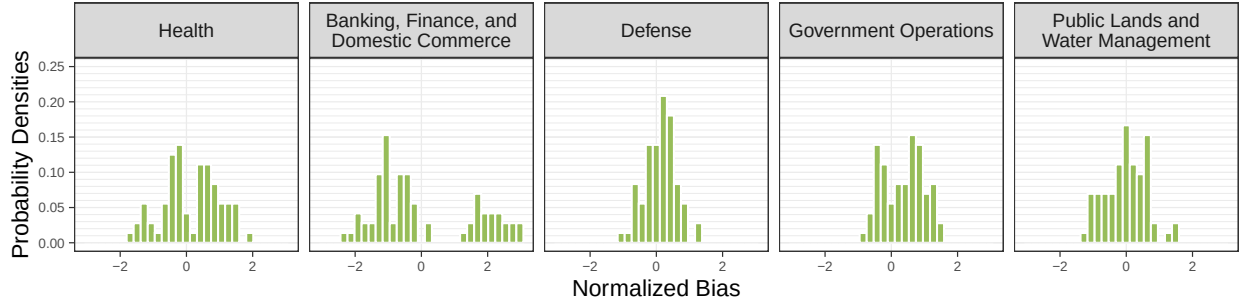
Note: The t-scores were obtained by fitting the model $V = Y^\top \beta = \sum_t Y_t \beta_t$ using 10,000 bills, with 95% confidence intervals computed via robust standard errors. Here, $V \in \{\text{Democrat}; \text{Senate}; \text{DW1}\}$ and $t \in \{\text{Health}; \text{Banking, Finance, and Domestic Commerce}; \text{Defense}; \text{Government Operations}; \text{Public Lands and Water Management}; \text{Other}\}$. The human-based coefficients β^* (red) were estimated using $Y_t^{\text{Human}} := 1\{Y^{\text{Human}} = t\}$, while LLM-based coefficients β^{LLM} (GPT-3.5 in green circles, GPT-4o in blue triangles) used $Y_{t,m,p}^{\text{LLM}} := 1\{Y_{m,p}^{\text{LLM}} = t\}$, with $m \in \{\text{GPT-3.5}, \text{GPT-4o}\}$ denoting the LLM model and $p \in \{1, \dots, 12\}$ representing the prompt. In each subplot, the t-score estimates were sorted in ascending order for clarity.

Figure 3: Bias Density of β_t^{LLM} for all t . Proxy on the RHS.



Note: The bias estimates were calculated as the difference between the LLM-based coefficient β_t^{LLM} and the human-based coefficient β_t^* , using a regression of the form $V = Y^T \beta = \sum_t Y_t \beta_t$, where $V \in \{\text{Democrat; Senate; DW1}\}$ and $t \in \{\text{Health; Banking, Finance, and Domestic Commerce; Defense; Government Operations; Public Lands and Water Management; Other}\}$. For human-based coefficients, the covariates are $Y_t^{\text{Human}} := 1\{Y^{\text{Human}} = t\}$, while for LLM-based coefficients, the covariates are $Y_{t,m,p}^{\text{LLM}} := 1\{Y_{m,p}^{\text{LLM}} = t\}$, with $m \in \{\text{GPT-3.5, GPT-4o}\}$ denoting the LLM model and $p \in \{1, \dots, 12\}$ representing the prompt. The human-based coefficient β_t^* was computed using the full 10,000 bills, whereas the LLM-based coefficient β_t^{LLM} was calculated as the average of $N = 1,000$ simulation runs, with each run's estimate based on a sample of 5,000 bills. The bias estimates are independent of the validation proportion η , though they were generated in the experiment run with $\eta = 5\%$.

Figure 4: Normalized Bias Density of β_t^{LLM} for all t . Proxy on the RHS.



Note: The normalized bias estimates were calculated as the difference between the LLM-based coefficient β_t^{LLM} and the human-based coefficient β_t^* , divided by the coefficient's sample standard deviation computed across $N = 1000$ samples. All other variables are as defined in the previous figure. The bias estimates are independent of the validation proportion η , though they were generated in the experiment run with $\eta = 5\%$.

Figure 5: Bias Density of β_t , for all t , by Model and Proportion of Validation Samples. Proxy on the RHS.

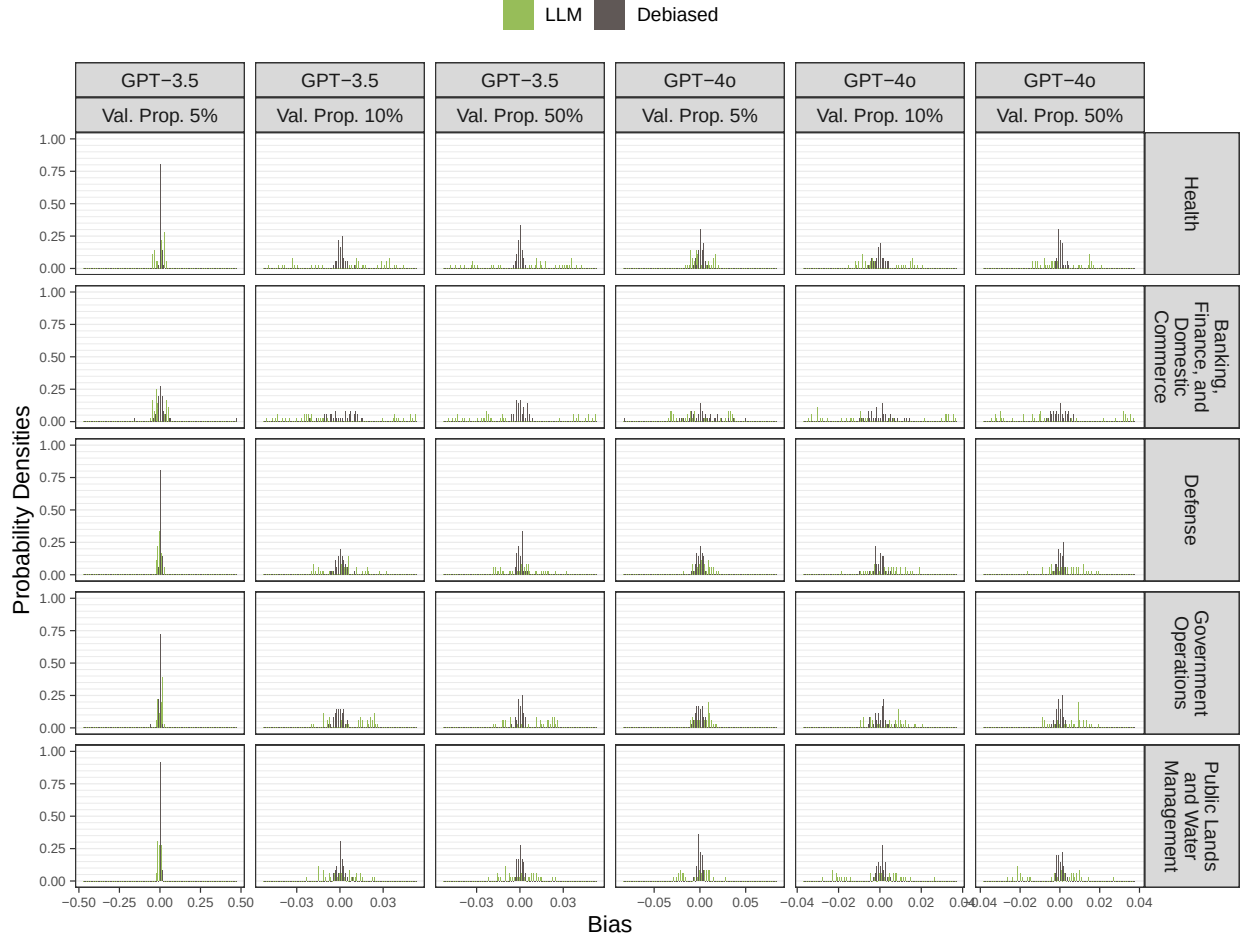


Figure 6: Normalized Bias Density of β_t , for all t , by Model and Proportion of Validation Samples. Proxy on the RHS.

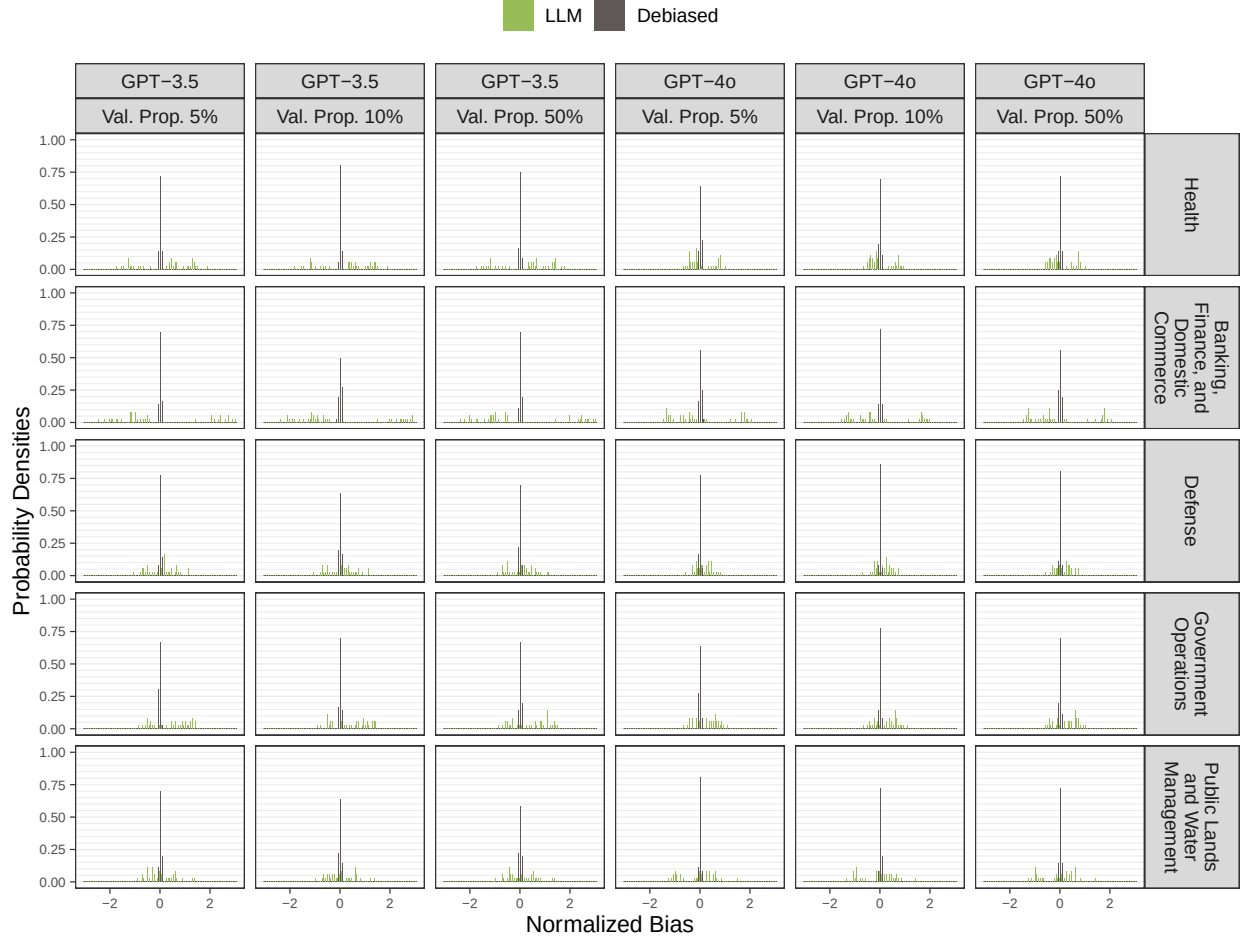


Figure 7: MSE Empirical CDF of β_t for all t by Proportion of Validation Samples. Proxy on the RHS.

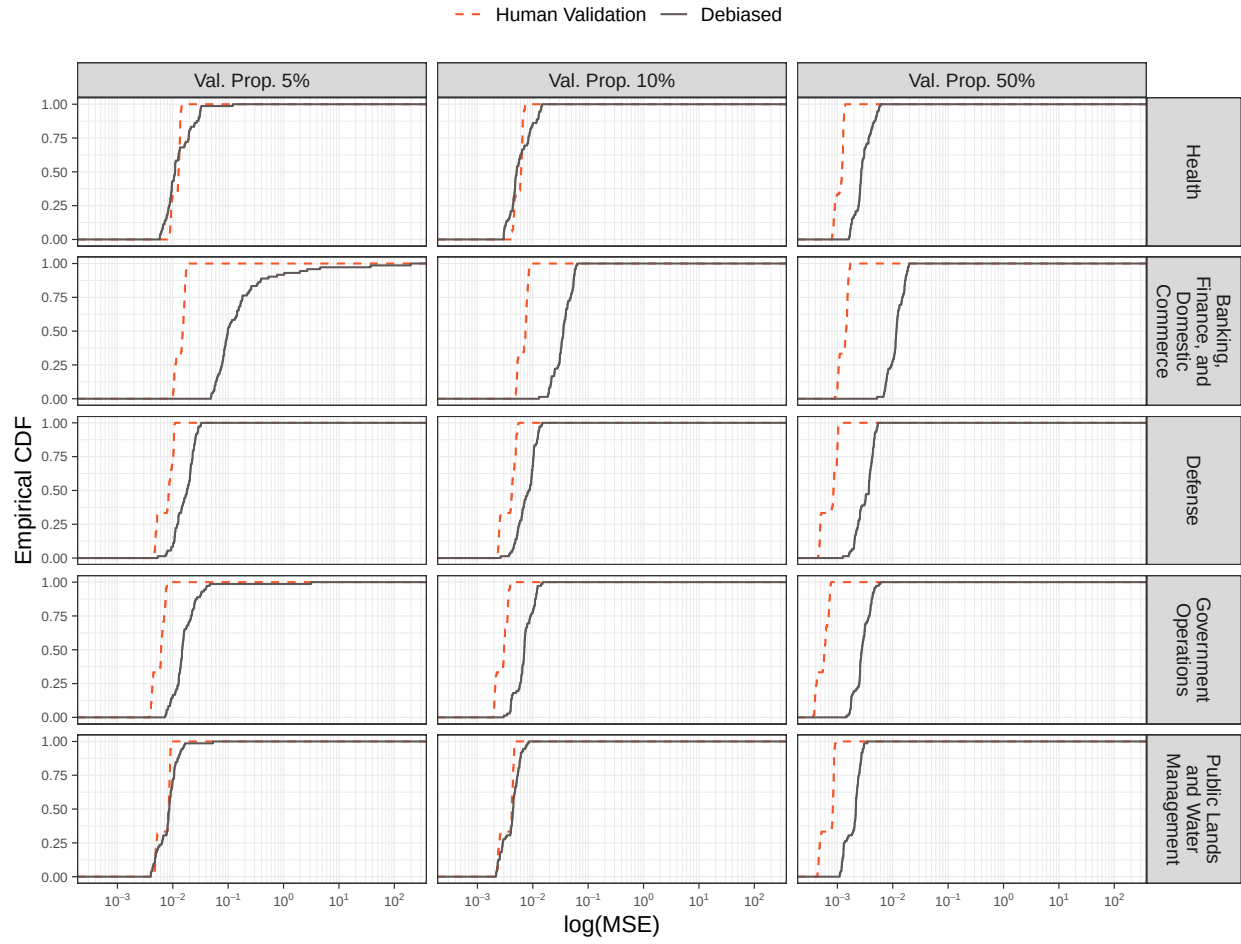


Figure 8: MSE Empirical CDF of β_t , for all t , by Model and Proportion of Validation Samples. Proxy on the RHS.

